

tf.compat.v1.math.exp Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/math/exp

Computes exponential of x element-wise. $y = e^x$.

Function Signatures

```
tf.compat.v1.math.exp( x, name=None )  
math.exp(x)  
x = tf.constant(2.0)  
tf.math.exp(x)
```

Code Examples

```
tf.compat.v1.math.exp(  
x, name=None  
)
```

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tf.compat.v1.math.exp(  
x, name=None  
)
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```
x = tf.constant(2.0)  
tf.math.exp(x)
```

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x = tf.constant(2.0)
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```
tf.math.exp(x)
```

```
x = tf.constant([2.0, 8.0])  
tf.math.exp(x)
```

```
x = tf.constant([2.0, 8.0])
```

Documentation

Computes exponential of x element-wise. $(y = e^x)$.

This function computes the exponential of the input tensor element-wise.

i.e. `math.exp(x)` or (e^x) , where x is the input tensor.

(e) denotes Euler's number and is approximately equal to 2.718281.

Output is positive for any real input.

For complex numbers, the exponential value is calculated as

$$\begin{aligned} & \\ e^{x+iy} &= \{e^x\} \{e^{iy}\} = \{e^x\} (\{\cos (y) + i \sin (y)\}) \\ & \end{aligned}$$

For $1+i$ the value would be computed as:

$$\begin{aligned} & \\ e^1 (\cos (1) + i \sin (1)) &= 2.7182817 \times (0.5403023+0.84147096i) \\ & \end{aligned}$$

Returns

numpy compatibility

Equivalent to `np.exp`